Embedded systems Architecture (BACSE324)

Assessment 10: Seven Segment Display Interfacing and Distance Measurement Using Arduino Uno

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Contents

Component	Page No.	Remarks
Program 1, 2	3, 9	
AIM of Experiment	3, 9	
Tools Required	3, 9	
Basic Brief Theory	3, 9	
Circuit Connections	4, 10	
Arduino Code with Comments	5, 11	
Calculations	6, 12	
Procedure to perform the AIM	6, 12	
Output Snips with name	7, 13	
Results	8, 14	
Inference	8, 14	

Signature of TRA

Program 1

Design a hardware circuit to interface a seven segment display with Arduino Uno. Display 0 to 9 and A to F, using 220 ohm resistors.

Aim of the Experiment

To interface a common-cathode seven segment display with Arduino Uno and display the sixteen hexadecimal symbols from 0 to F in sequence.

Tools Required

- Arduino Uno R3
- One common-cathode seven segment display
- Seven 220 ohm resistors, one per segment
- Breadboard and jumper wires
- USB cable
- Computer with Arduino IDE

Brief Theory

A seven segment display contains LED segments labelled a to g. Different combinations represent decimal digits and hexadecimal symbols. For a common-cathode display, connect the common terminals to GND; a HIGH output lights a segment through its own 220 ohm resistor.

Assign segments a to g to D2 to D8 in order. A sixteen-entry lookup table defines the segment patterns for 0 to 9 and A to F. Show each pattern for about one second before advancing. Lowercase b and d are used because uppercase B and D cannot be represented distinctly on seven segments.

Circuit Connections

Component terminal	Connection
Display segment a	Arduino D2 through one 220 ohm resistor
Display segment b	Arduino D3 through one 220 ohm resistor
Display segment c	Arduino D4 through one 220 ohm resistor
Display segment d	Arduino D5 through one 220 ohm resistor
Display segment e	Arduino D6 through one 220 ohm resistor
Display segment f	Arduino D7 through one 220 ohm resistor
Display segment g	Arduino D8 through one 220 ohm resistor
Common-cathode terminals	Arduino GND; verify the display datasheet pinout
Decimal point	Leave disconnected
Arduino USB port	Computer USB port

Use segment labels rather than assuming physical pin numbers. Display pinouts vary by part. This connection plan is for a common-cathode display; a common-anode part requires different wiring and inverted logic.

Display Patterns

Symbol	Segments ON
0	a b c d e f
1	b c
2	a b d e g
3	a b c d g
4	b c f g
5	a c d f g
6	a c d e f g
7	a b c
8	a b c d e f g
9	a b c d f g
A	a b c e f g
b	c d e f g
C	a d e f
d	b c d e g
E	a d e f g
F	a e f g

Arduino Code with Comments

Manual calculation:

None

Procedure

1. Identify the common-cathode display and locate segments a to g using its datasheet. Disconnect power before wiring.
2. Connect the common cathodes to GND. Connect each segment to D2 to D8 through its own 220 ohm resistor, following the connection table.
3. Prepare the Arduino sketch in the reserved code section. Set the seven segment pins as outputs and initially turn every segment OFF.
4. Create patterns for all sixteen symbols using the Display Patterns table. Apply HIGH to active segments and LOW to inactive segments.
5. Display each symbol for about one second, then advance through 0 to 9 and A, b, C, d, E, F. Repeat the sequence continuously.
6. Select Arduino Uno and the correct port, verify the sketch and upload it.
7. Check each symbol against the pattern table. Correct wiring errors with power disconnected and capture clear output photographs.

Output Snips

Result

Expected result: The display should show 0 to 9 followed by A, b, C, d, E and F, then repeat. Each symbol should use the correct segment pattern. Verify the sequence on the connected circuit.

Inference

Seven independently controlled LED segments can represent sixteen hexadecimal symbols using a lookup table. One resistor per segment limits current. Correct common-terminal wiring and segment mapping are essential for accurate characters.

Program 2

Design a distance-measuring hardware circuit with six different LEDs and a 16 by 2 LCD. Show the measured distance on the LCD and switch ON the LED corresponding to that distance.

Aim of the Experiment

To measure distance using an HC-SR04 ultrasonic sensor, show the measured distance in centimetres on a 16 by 2 LCD and select one of six LEDs according to the measured range.

Tools Required

- Arduino Uno R3
- HC-SR04 ultrasonic distance sensor
- 16 by 2 LCD compatible with the LiquidCrystal library
- 10 kilohm potentiometer for LCD contrast
- Six LEDs of different colours
- Six 220 ohm resistors for the LEDs
- 220 ohm LCD backlight resistor if required by the module
- Breadboard and jumper wires
- USB cable and computer with Arduino IDE
- Ruler or measuring tape and a flat target

Brief Theory

The HC-SR04 emits an ultrasonic burst after a 10 microsecond trigger pulse. Its ECHO pulse duration represents the outward and return travel time of the sound. Distance is calculated from this duration and displayed on the LCD; a ruler provides an independent comparison.

Use a parallel LCD in four-bit mode with the LiquidCrystal library. Compare each valid reading with six non-overlapping distance bands and light only the selected LED. The thresholds below are demonstration choices because the question does not specify them. If no valid echo is received, display No echo and turn all six LEDs OFF.

Circuit Connections

Component terminal	Connection
HC-SR04 VCC and GND	Arduino 5 V and GND
HC-SR04 TRIG and ECHO	Arduino D2 and D3 respectively
LED 1 to LED 6 anodes	D4, D5, D6, D7, D8, D9 respectively; each through its own 220 ohm resistor
All six LED cathodes	Arduino GND
LCD VSS, VDD and RW	GND, 5 V and GND respectively
LCD VO contrast	10 kilohm potentiometer wiper; outer terminals to 5 V and GND
LCD RS and Enable	Arduino D10 and D11 respectively
LCD D4 and D5	Arduino D12 and D13 respectively
LCD D6 and D7	Arduino A0 and A1 used as digital outputs
LCD D0 to D3	Leave disconnected in four-bit mode
LCD backlight A and K	5 V through required current-limiting resistor, and GND; check module specifications
Arduino USB port	Computer USB port

Use LiquidCrystal lcd(10, 11, 12, 13, A0, A1) for this pin plan. Join every component ground to Arduino GND. Label the six LEDs 1 to 6 so that their distance bands remain clear regardless of the colours available.

LED Distance Bands

Selected LED	Measured distance in cm
LED 1	2 to less than 10
LED 2	10 to less than 20
LED 3	20 to less than 30
LED 4	30 to less than 40
LED 5	40 to less than 50
LED 6	50 to 400
All LEDs OFF	No echo or reading outside 2 to 400

Arduino Code with Comments

Manual calculation:

Distance in cm = echo duration in microseconds x 0.0343 / 2. The factor 0.0343 cm per microsecond approximates the speed of sound at room temperature; division by two accounts for the return journey.

Example only: for an echo duration of 1166 microseconds, distance = 1166 x 0.0343 / 2 = 20.00 cm. This falls in the LED 3 band. Compare measured readings with a ruler; this example is not an experimental observation.

Procedure

1. Disconnect power and wire the ultrasonic sensor, six LEDs and LCD according to the connection table. Use a separate resistor for each LED.

2. Connect USB power, adjust the LCD contrast potentiometer and initialise the LCD as 16 columns by 2 rows.

3. Prepare the sketch in the reserved section. Configure TRIG as OUTPUT, ECHO as INPUT and the six LED pins as OUTPUT. Start with all LEDs OFF.

4. Send a 10 microsecond trigger pulse. Measure the ECHO pulse with a bounded timeout, such as 30000 microseconds, and calculate distance in centimetres.

5. For each valid reading, turn all LEDs OFF and then turn ON only the LED for its distance band. Treat missing or out-of-range readings as invalid.

6. Show the distance on LCD row 1, for example Distance: 20.0cm, and the selected LED on row 2, for example LED: 3. Clear unused character positions when a value becomes shorter.

7. Verify and upload the sketch. Place a flat target at several ruler-measured positions, including both sides of each distance threshold, and allow about 100 milliseconds between readings.

8. Compare the displayed distance with the ruler reading. Check that exactly one LED lights for a valid distance, then capture the LCD and corresponding LED in the output photographs.

Output Snips

Result

Expected result: The LCD should update the measured distance in centimetres, and exactly one of six LEDs should indicate its band. Moving the target across a threshold should change the selected LED. An invalid reading should show No echo or Out of range with all LEDs OFF.

Inference

Ultrasonic echo timing provides a distance estimate, while the LCD and six LEDs present numeric and range-based outputs. Accuracy depends on target alignment, surface and ambient conditions. Comparing readings with a ruler helps assess the measurement error.

ESA Lab Assessment report

Sl.no.	Components	Marks Obtained
1	Pre Lab Work (Flow chart and Algorithm) documentation (2)	
2	Hardware or software Implementation (2)	
3	Manual calculation , Output data table / graph verification (2)	
4	Result with Inference (2)	
5	Post Lab Work (C Code with comments) documentation (2)	
	Total (10)	

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